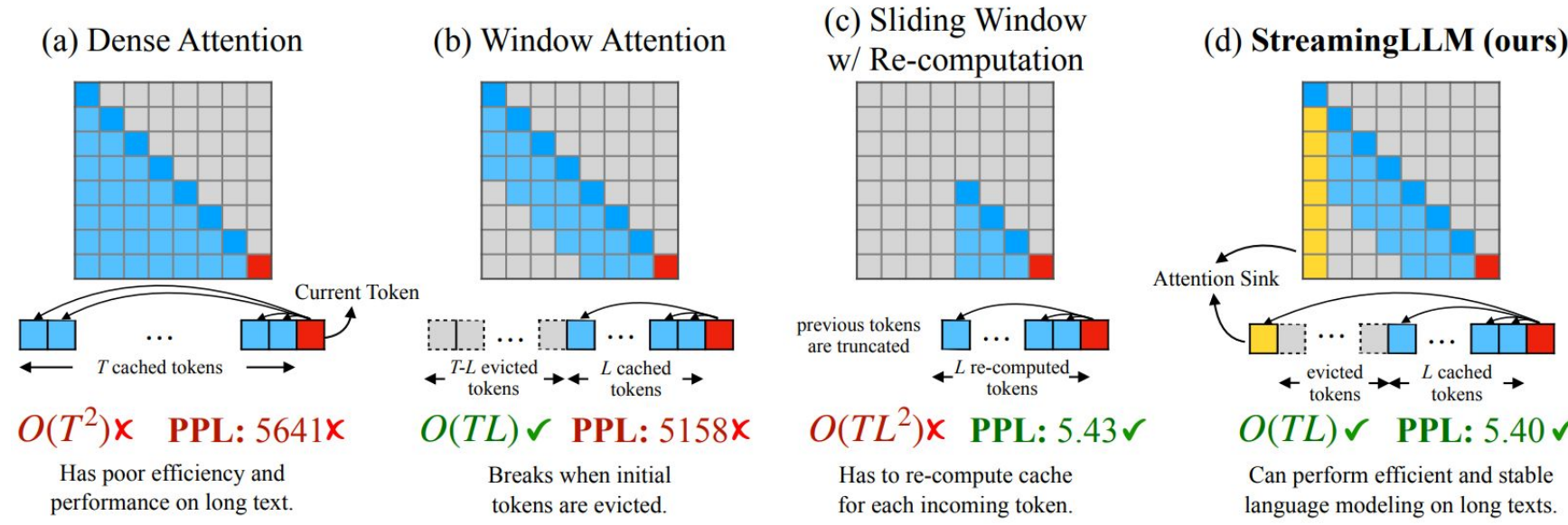


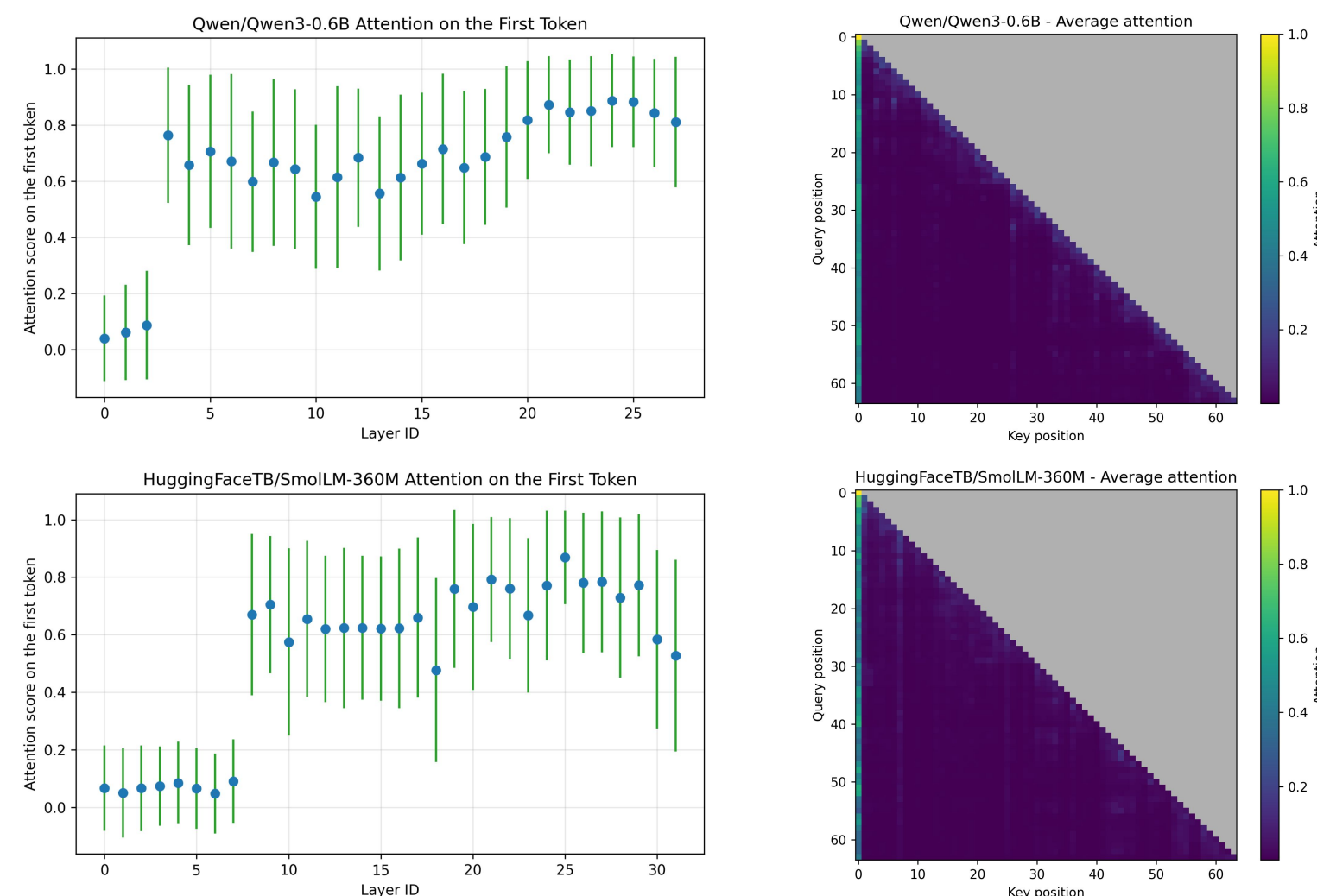


Motivation



Intuition: Early tokens absorb attention mass due to softmax normalization. Keeping them stabilizes attention pattern even when most of distant history is removed.

Evidence



The main claim of the paper is about **quality under long streams**. Once generation exceeds the active window:

- Sliding window attention fails, with a sharp NLL jump.
- Sliding window with re-computation remains stable.
- Attention sinks stay close to the re-computation baseline.

Experimental setup

Dataset: PG19

Model	Context Tokens	Windows
smoLLM	2k	20k
Qwen	32k	100k

Metrics:

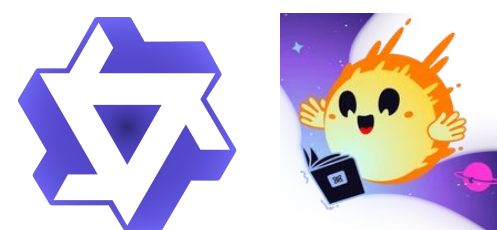
- Perplexity / NLL
- Peak memory
- Average token latency

The experiments compare both **asymptotic claims** and real system behaviour under long-context inference.

Note: Some Qwen3 experiments were run on 20k token with a 5k window

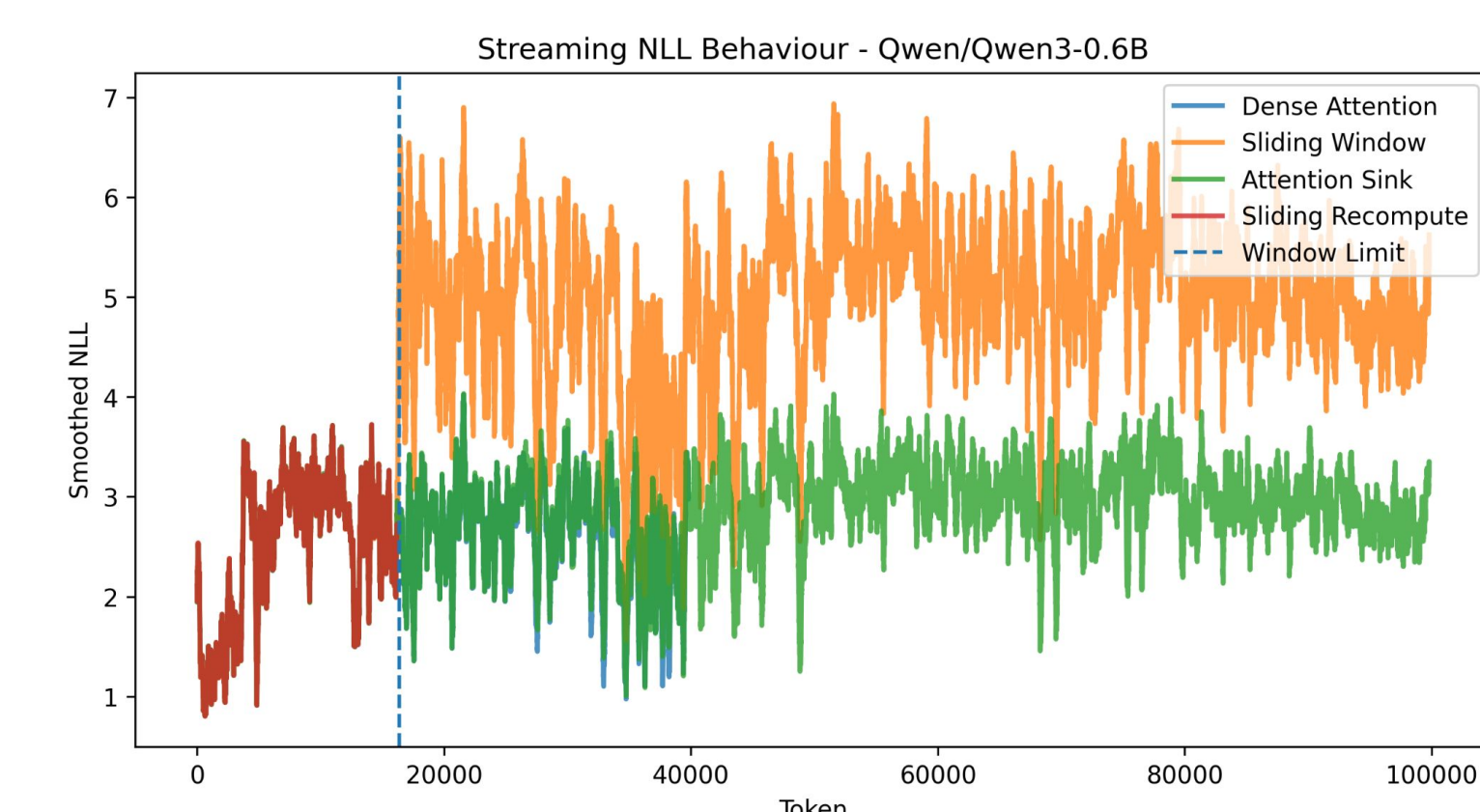
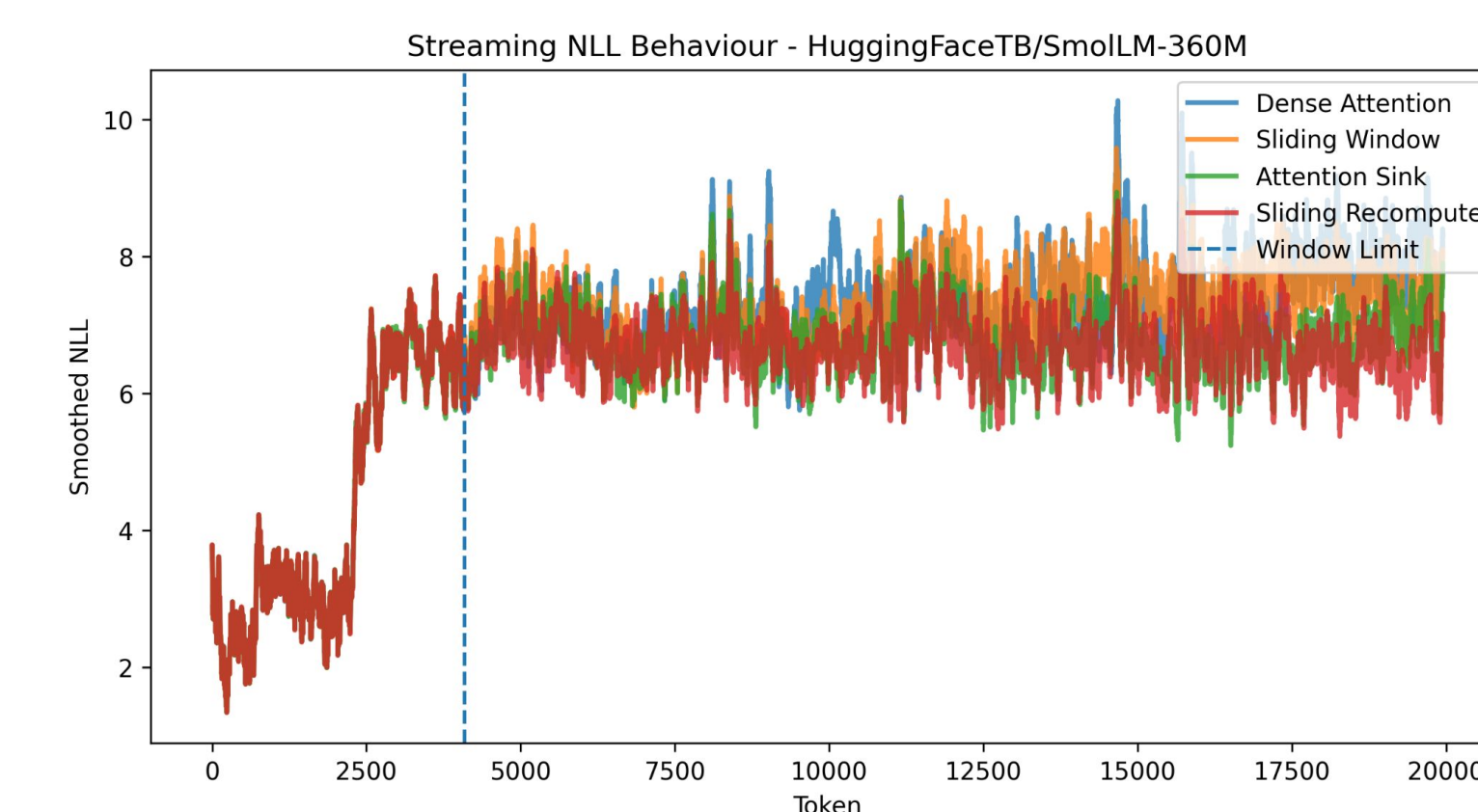
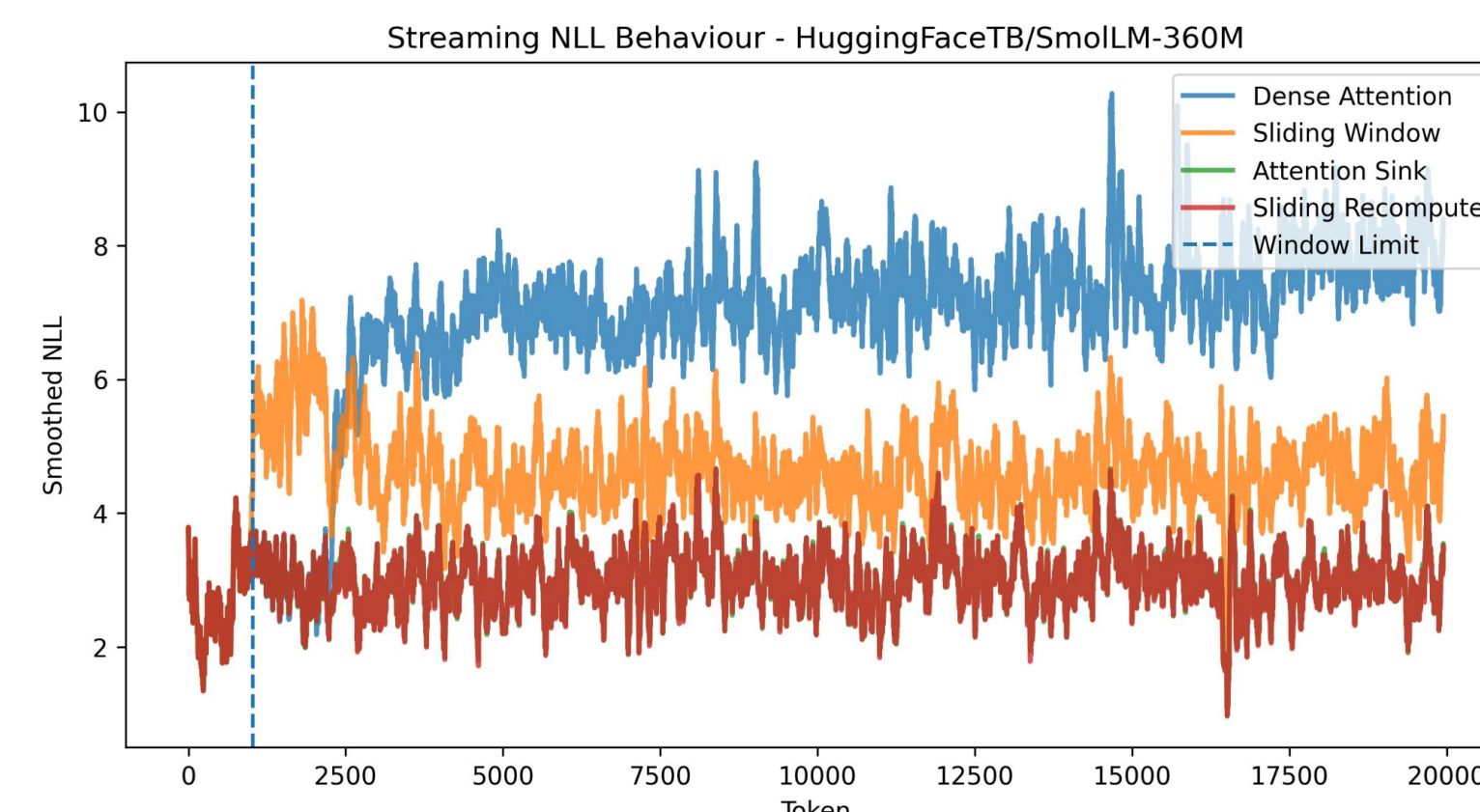
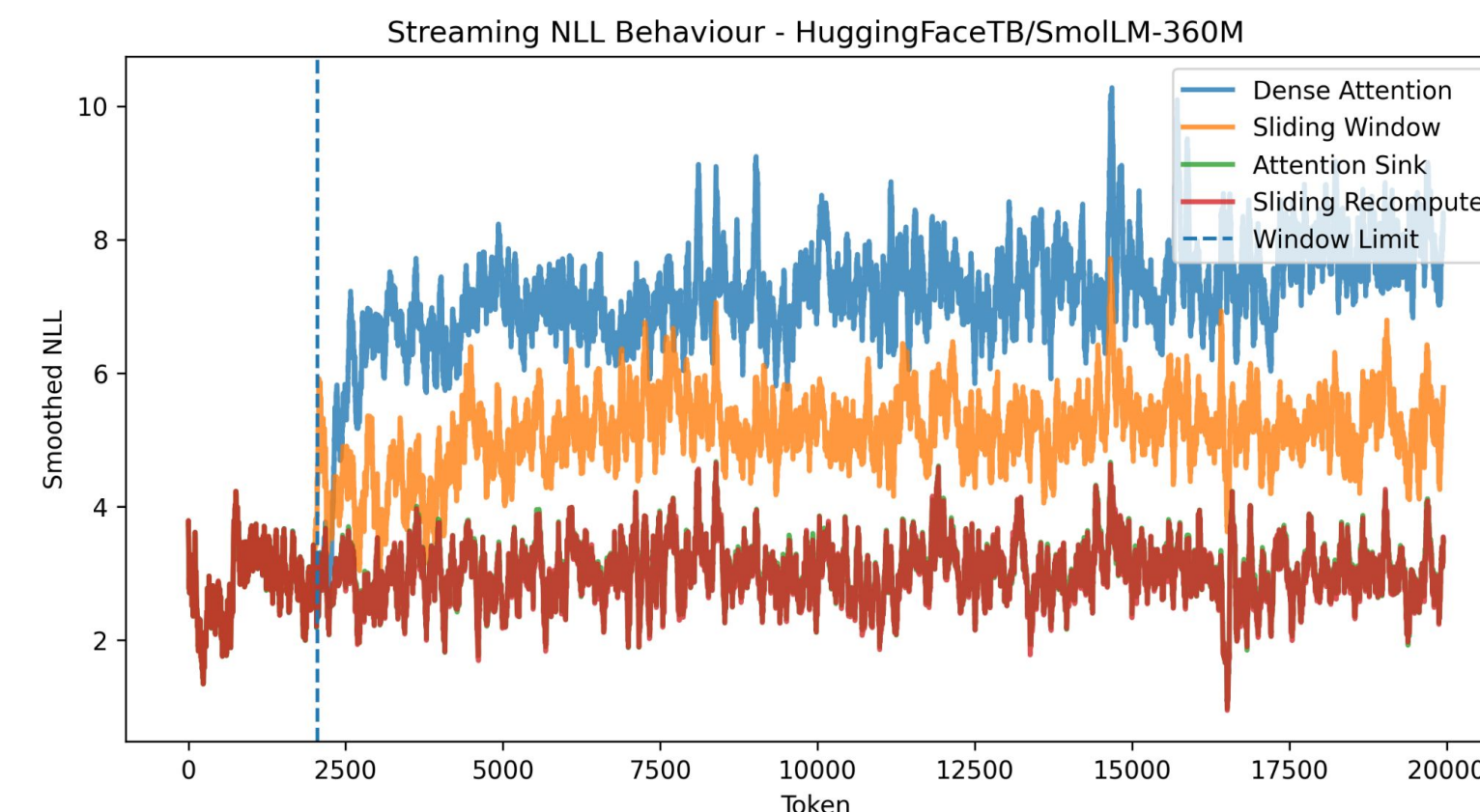
Choice of models

- Newer architecture vs Llama-2 based
- 32k vs 2k context length
- 1000k vs 10k base RoPE period

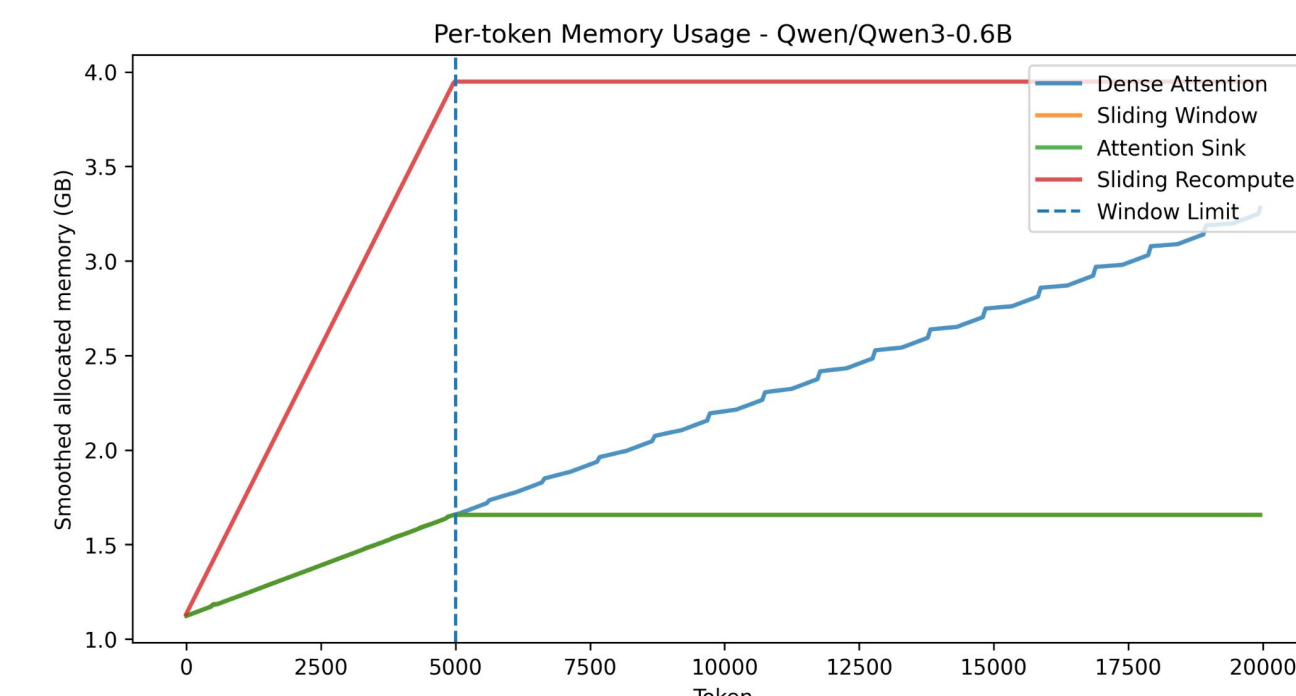


Perplexity over tokens

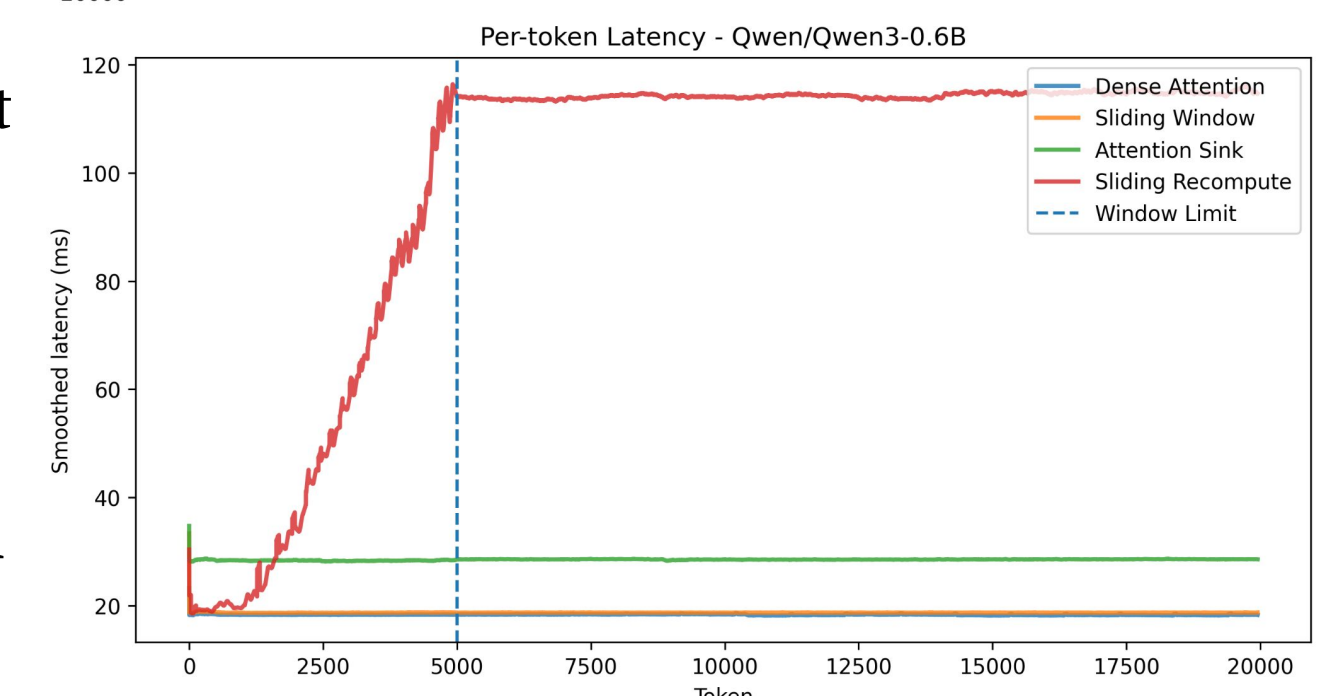
- Context Mismatch in Sliding Windows
- Recomputation is Too Slow
- Attention Sinks Solve Both



Memory and Latency



- AS >> SW > DA. All three have constant latencies
- SR latency increases as the window is filled with tokens. With a full 5k window, SR is much slower than AS, SW and DA
- With a 16k window, SR latency is even higher



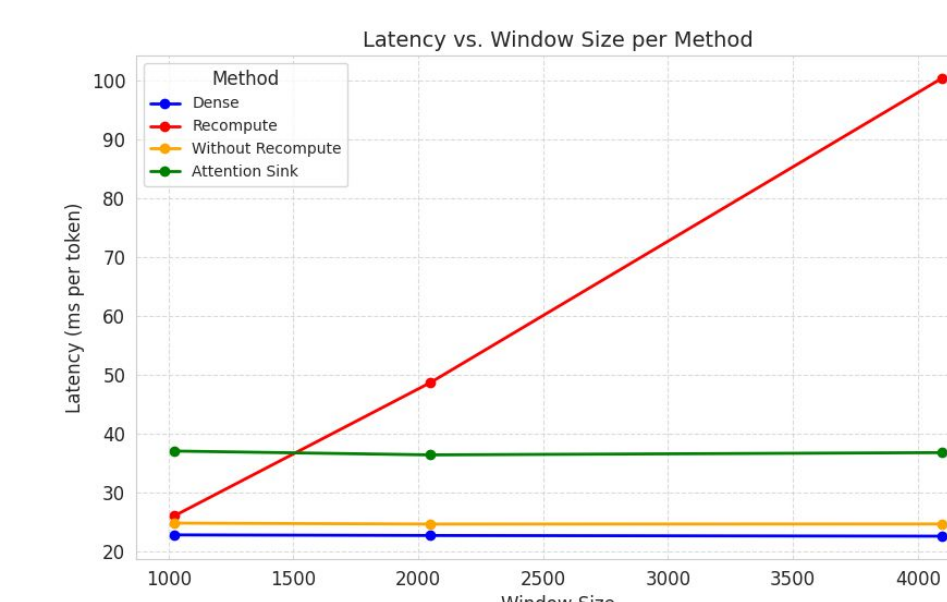
Practical Caveats

Perplexity of dense attention behaves differently across models:

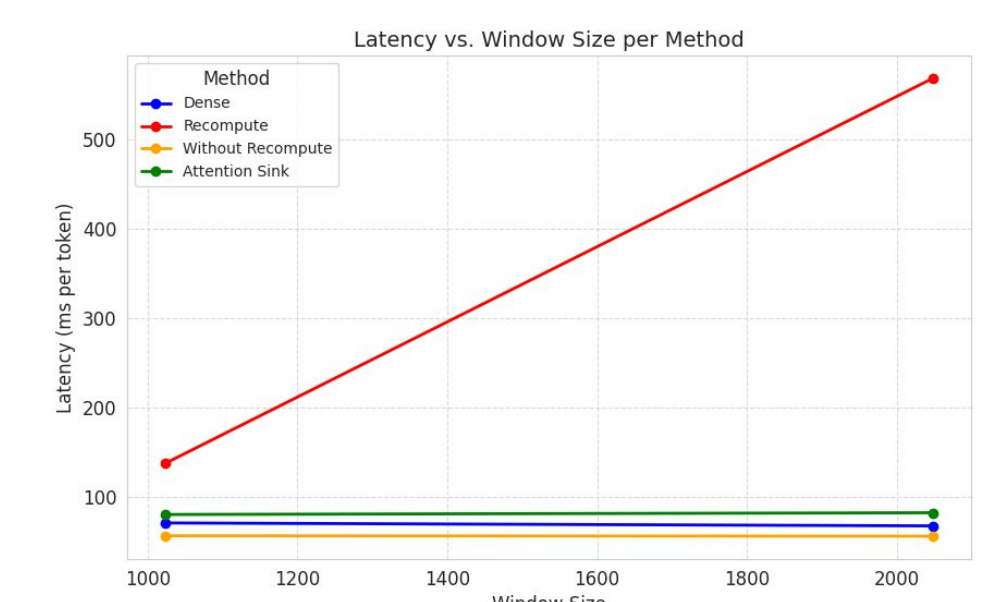
- **smoLLM:** degrades beyond the model context length (as in the original paper)
- **Qwen:** remains stable beyond the model context length: testimony to the improvement of more recent LLMs models

Non-asymptotically, system implementation can have a large impact on efficiency: latency of dense attention and attention sinks, memory of sliding recompute, impact of system changes

smoLLM-360M, rtx 5060ti + linux



smoLLM-360M, rtx 4060 + windows



Takeaway message

- The claim the paper makes about attention sinks offering the optimal performance (measured by perplexity) while having the optimal asymptotic efficiency (in terms of time and memory complexity) holds true
- In practice, GPU and system optimization of the attention mechanisms also matters a lot. For sequences that are not extremely long (~20k tokens), these matter more than asymptotic behaviour
- With newer models (Qwen3-0.6B), dense attention doesn't lose performance when exceeding the models training length (where as it did with older models, such as those based on llama-2)